

Experimental Evaluation and Hardware Overhead Analysis of Zero-Blocking Optimization in a 4x4 Tensor Grid Accelerator

Objective

To quantitatively evaluate the operation of the *zero-Blocking circuit* in a 4×4 matrix multiplication accelerator by measuring the circuit's switching activity as a function of input data sparsity.

Methodology & Experimental Setup

- **Simulation Tools:** Icarus Verilog for RTL compilation/execution, coupled with a custom C-based **IEEE VPI (Verilog Procedural Interface)** monitoring module (activity.c) registered directly into the simulation runtime.
- **Evaluation Metric:** The total number of signal value changes monitored strictly at the logic netlist level during active instruction execution. This provides an exact indicator of dynamic power consumption.
- **IEEE VPI Activity Isolation:** The VPI C-module isolates the netlist toggles per test execution step, eliminating testbench noise floor overhead and yielding true hardware-level switching metrics.
- **Test Scenarios:** Executed via zero_block_eval_tb.v (make eval), consisting of a dense baseline workload followed by progressive tile-level zero-blocking evaluations (0 to 3 zeroed-out 2×2 Tensor Tiles, corresponding to 0% to 75% matrix sparsity).

Execution Trace Snippet

```
VCD info: dumpfile tb_tensor.vcd opened for output.  
SWITCHING ACTIVITY: 497 Toggles  
TEST 1: Signed Tensor MAC Result (A x B + C = D)
```

$$A = \begin{bmatrix} 2 & -1 & 1 & 0 \\ -2 & 3 & 2 & -5 \\ -4 & 1 & 1 & 1 \\ 0 & 2 & -1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & -2 & 1 \\ 3 & -1 & 0 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & -4 & 3 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} -10 & 10 & 0 & 3 \\ 5 & -2 & 4 & 1 \\ -16 & 2 & 2 & -6 \\ 1 & 5 & 3 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} -9 & 15 & -3 & 5 \\ 11 & 11 & -5 & 4 \\ -14 & -11 & 14 & -5 \\ 7 & -5 & 8 & 4 \end{bmatrix}$$

```
SWITCHING ACTIVITY: 1106 Toggles  
TEST 2: Zero Blocking ON - Matrix A with 1 Zero-Tile (A00)
```

$$A = \begin{bmatrix} 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & -2 \\ 1 & 4 & -3 & 2 \\ 2 & 1 & 1 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 & 1 & 2 \\ -2 & 3 & 0 & 1 \\ 3 & -3 & 1 & 0 \\ 1 & 2 & -2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 7 & -4 & 0 & 1 \\ 7 & -13 & 7 & -2 \\ -13 & 26 & -6 & 8 \\ 8 & 8 & -3 & 8 \end{bmatrix}$$

SWITCHING ACTIVITY: 1746 Toggles

TEST 3: Zero Blocking ON - Matrix A with 2 Zero-Tiles (A00, A01)

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 2 \\ -2 & 3 & 4 & -3 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 & 1 & 2 \\ -2 & 3 & 0 & 1 \\ 3 & -3 & 1 & 0 \\ 1 & 2 & -2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 7 & 6 & -1 & 7 \\ -1 & -11 & 8 & -4 \end{bmatrix}$$

SWITCHING ACTIVITY: 2149 Toggles

TEST 4: Zero Blocking ON - Matrix A with 3 Zero-Tiles (A00, A01, A10)

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & -2 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 & 1 & 2 \\ -2 & 3 & 0 & 1 \\ 3 & -3 & 1 & 0 \\ 1 & 2 & -2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 11 & -5 & -1 & 2 \\ -5 & 8 & -4 & 1 \end{bmatrix}$$

SWITCHING ACTIVITY: 2484 Toggles

zero_block_eval_tb.v:280: \$finish called at 195 (1s)

Analysis & Benchmark Results

By decoupling cumulative activity measurements, the isolated execution cost (Δ Toggles) for each individual workload is extracted:

Test Scenario	Sparsity Level	Zero-Blocking Status	Cumulative Toggles	Execution Cost	Dynamic Power Reduction / Dense Baseline
Initial Reset & Setup	-	OFF	497	497	-
TEST 1 (Dense Matrices)	0%	OFF	1,106	609	0.0%
TEST 2 (1 Zero Tile - A_{00})	25%	ON	1,746	640	+5.1%
TEST 3 (2 Zero Tiles - A_{00}, A_{01})	50%	ON	2,149	403	-33.8%
TEST 4 (3 Zero Tiles - A_{00}, A_{01}, A_{10})	75%	ON	2,484	335	-45.0%

Technical Observations

- Isolation of Real Hardware Gains:** An in-memory VPI event monitor reveals that the Zero-Blocking optimization delivers up to **45.0% dynamic switching activity reduction** at 75% sparsity.
- Control Logic Overhead at Low Sparsity:** At 25% sparsity (1 Zero Tile), execution cost increases slightly from 609 to 640 toggles (+5.1%). This reflects the fixed activation overhead of the zero-detection comparators and gating multiplexers.
- High Sparsity Scaling:** Once sparsity reaches $\geq 50\%$, the power saved by gating entire multiplier-accumulator (MAC) sub-trees far outweighs the detection logic cost, reducing execution activity down to **335 toggles** at 75% sparsity.

Conclusions

- High Efficiency in Sparse Workloads:** The Zero-Blocking mechanism is highly effective, delivering up to a **45% reduction in dynamic switching activity** during matrix multiplication.
- Architectural Trade-Off:** While single-tile gating exhibits a minor control logic overhead (~5%), the architecture provides steep power savings for medium-to-high sparsity workloads, making it ideal for sparse neural network acceleration.